

MYZONE – SFRC 360

Unified Smart Campus Experience, Institutional Intelligence & Campus Operations Platform

Complete Product Requirements • SFRC Institutional Data • CivicFix Integration • Pragma AI • Production Technology Architecture

Prepared: 27 September 2026

Reference institution: The Standard Fireworks Rajaratnam College for Women (SFRC), Sivakasi

Document purpose: serve as the single source of truth for building MyZone in an AI-assisted development environment such as Google Antigravity. The document separates public institutional facts, private/operational data, AI capabilities, integration boundaries, and production engineering concerns.

1. Executive Summary

MyZone – SFRC 360 is not intended to replace SFRC’s existing ERMS. It is a unified smart-campus experience and intelligence layer that brings student/faculty/parent services, campus operations, institutional knowledge, mentoring, events, career/research/student-life functions, AI assistance and maintenance workflows into one responsive web platform.

SFRC already operates a cloud-based ERMS covering online admission, fee payment, attendance, OBE/CBCS, examination, HR & Payroll, Registrar Office, Hostel Management, Alumni Management, Feedback Management and automated mailing. The official Best Practices page also describes faculty, student, parent and examination portal capabilities. Therefore MyZone should be integration-ready rather than claiming to replace or directly connect to ERMS without an authorized API/data agreement. ■cite■turn0search1■turn0search3■

The two major differentiators are **Pragya AI** and **CivicFix / Campus Care**. Pragya is the permission-aware institutional assistant. CivicFix is the AI-assisted complaint and maintenance workflow: complaint intake → AI understanding → duplicate detection → routing/assignment → SLA/escalation → resolution → feedback and analytics.

The recommended production stack is **Next.js + TypeScript + Tailwind CSS + shadcn/ui + FastAPI + PostgreSQL/Supabase + pgvector + Pragya AI**, with Git/GitHub and a production deployment split between a web frontend host and backend/database services. This stack is modern, scalable, AI-friendly and practical for AI-assisted development.

Core positioning

Dimension	MyZone position
Institutional role	Unified digital campus experience / intelligence layer
Existing system relationship	ERMS-aware and integration-ready; not a claimed ERMS replacement
Primary users	Student, Faculty, Parent, Admin; operational staff via permission-based capabilities
AI	Pragya AI for grounded campus assistance and workflow initiation
Operations	CivicFix / Campus Care for maintenance and complaints
Delivery	Responsive production web app / PWA-oriented architecture
Development model	AI-assisted / vibe coding using Google Antigravity, with human review and testing

2. SFRC Institutional Baseline — Official Website Findings

The following baseline is derived from the current official SFRC website and linked institutional pages. Dynamic counts and live event data should be seeded into the database and displayed dynamically rather than hard-coded.

Institutional fact	Source-derived detail	MyZone use
Institution	The Standard Fireworks Rajaratnam College for Women, Thiruthangal Road, Sivakasi – 626123, Tamil Nadu, India.	Institution profile / footer / public info
Established	1968 (institutional history shown on the official site).	About / institutional timeline
Current homepage counters	Homepage currently presents 3436 Students, 160 Staffs, 19 Departments and 40 Programmes.	Seed/reference metrics only; use dynamic DB counts
Autonomy / OBE	College describes autonomous status from 2005 and Outcome-Based Education/CBCS processes.	Examination + OBE/attainment modules
Current quality milestone	2026 milestone: ISO 9001:2015 certificate for quality educational services.	Institution / quality dashboard
Recent academic expansion	2024: B.Sc. Data Science; 2025: B.Sc. Geography (SF); 2026 milestone includes current quality certification.	Programme master data

Institutional fact	Source-derived detail	MyZone use
Alumnae	Official milestones state an Alumnae portal was launched in 2022.	Alumnae module / future integration

Important data-governance note: the homepage counters and the number of named department/programme records can differ because the site is a live institutional portal with multiple categories and historical/current records. MyZone should derive counts from active database records and show “current as of” timestamps where appropriate.

3. Existing SFRC Digital Ecosystem and the MyZone Boundary

SFRC’s official Best Practices page describes digital transformation through a cloud-based ERMS, E-Content Development Cell and Information Resource Centre. ERMS includes online admission, fee payment, attendance, OBE/CBCS, examination, HR & Payroll, Registrar Office, Hostel Management, Alumni Management, Feedback Management and Automated Mailing. Faculty, student, parent and examination portals support role-specific functions. ■cite■turn0search1■

This makes MyZone more credible as an **experience + intelligence + operations layer**. It should own data that MyZone itself creates, and use adapters/interfaces for future authorized synchronization with institutional systems.

Source-of-truth model

Data class	Authoritative source	MyZone behavior
Public institutional facts	SFRC official website / approved documents	Index into institutional knowledge/RAG and public pages
Existing ERMS/private academic data	SFRC ERMS, if authorized	Integration adapter; never assume API access
MyZone operational records	MyZone PostgreSQL	Own and manage directly
CivicFix complaints	MyZone PostgreSQL + Storage	Own workflow, audit trail and SLA
Pragya answers	Authorized DB queries + approved RAG sources	Grounded response with source context
Sensitive data	Role-authorized systems	RLS/RBAC; never expose via unrestricted LLM context

4. Academic and Institutional Data to Seed

The official regular-programme page lists the following current regular programmes. ■cite■turn1search5■

Regular UG: B.A. Tamil; B.A. English; B.A. History; B.Com.; B.Com. (Computer Applications); B.Sc. Mathematics; B.Sc. Physics; B.Sc. Chemistry; B.Sc. Botany; B.Sc. Computer Science.

Regular PG: M.A. English; M.Sc. Mathematics; M.Sc. Chemistry.

The current self-financing page lists UG: B.Com.; B.Com. (Computer Applications); B.Com. (Professional Accounting); B.B.A.; B.Sc. Microbiology; B.Sc. Nutrition and Dietetics; B.Sc. Costume Design & Fashion; B.C.A.; B.Sc. Computer Science (SF); B.Sc. Psychology; B.Sc. Data Science; B.Sc. Geography. PG: M.A. Tamil; M.A. History; M.Com. (Computer Applications); M.Com. (Banking Technology); M.Com. (Human Resource Development); M.Sc. Physics; M.Sc. Botany; M.Sc. Microbiology; M.Sc. Computer Science; M.C.A. PhD: Tamil, Commerce, English, Chemistry, Mathematics, History, Physics and Botany. ■cite■turn1search7■

Certificate programmes currently listed: Certificate in Thollezhuthiyal; Certificate in Gandhian Thought; Certificate in Journalism and Mass Communication; Certificate in French; Certificate in R Programming; Certificate in Data Analysis for Statistics; Certificate in MATLAB; Certificate in Instrumentation Techniques; Certificate in Pyrotechnic Chemistry; Certificate in Forensic Science; Certificate in Cheminformatics; Certificate in Mushroom Farming; Certificate in Women’s Outfit; Certificate in Medical Coding; Certificate in Interior Decoration; Certificate in Chinese Cuisines; Certificate in Mushroom Cultivation; Certificate in Home Decors; Certificate in Big Data Computing;

Certificate in Data Entry Operations (DEO); Certificate in Desktop Publishing (DTP); Certificate in 2D Animation; Certificate in Office Automation; Certificate in Digital Marketing; Certificate in Service Marketing; Certificate in Professional Data Analysis; Certificate in Event Management; Certificate in Museum Studies; Certificate in Yoga; Certificate in Library and Information Science; Certificate in Behavior Modification; Certificate in AI & ML practice; Certificate in Vermicomposting Technology. ■cite■turn1search12■

Diploma / part-time diploma examples listed on the official programme pages: Diploma in Gandhian Thought; Diploma in Medical Laboratory Technology; Diploma in 3D Modeling and Animation; Diploma in Computer Applications (DCA); part-time Diploma in Costume Design & Fashion; Jute and Eco Products Design; Banking and Pastries; Communication & Functional English; Generative AI.

Skill / YWED ecosystem: Typewriting, Tailoring, Shorthand, COA, Jute Products Making; short-term skills such as Aari work, Lippan Art, Hand Embroidery, Brooch Embroidery, Arathi Plate Making, Millet Snacks Making, Smocking Cushions Making, Beautician Course, Veena and String Art; YWED activities include Basics of Bakery, Cuisines of India, Food Preservation, Blouse Designing, Hoop Art, Saree Embellishment, Surface Embellishment, Handicrafts, Painting, Doll Making, Herbal Cosmetics, Fabric Bag Making, Digital Photo Print and E-services. The official YWED page says students are expected to complete at least one long-term course or two short-term skill courses in an academic year. ■cite■turn1search3■

5. Complete MyZone Product Scope — A to Z

Module	Production scope
A. Identity, Authentication & RBAC	Login, session management, profile, role/capability mapping, password reset, device/session controls, audit trail. Roles: Student, Faculty, Parent, Admin; operational permissions for Mentor, Maintenance, Librarian, Hostel Warden, Placement, IQAC and COE without creating unnecessary top-level roles.
B. Student Dashboard	Personalized summary of attendance, academic performance, timetable, upcoming events, assignments, mentor, complaints, library due items, placement opportunities, notifications and Pragma.
C. Faculty Dashboard	Teaching schedule, assigned courses, attendance entry, mark entry, syllabus/resources, mentee list, mentoring notes, events, complaints and relevant analytics.
D. Parent Dashboard	Ward profile, authorized attendance/progress, mentor updates, notices/events, relevant requests and notifications.
E. Admin / Institutional Dashboard	Users, departments, programmes, academic calendar, events, complaints, maintenance, facilities, announcements, reports, audit logs, configuration and operational analytics.
F. Academics	Departments, programmes, batches, semesters, courses, course enrollment, timetable, attendance, marks, assignments, exams, results, OBE/CO/PO, learning resources.
G. Learning & E-Content	Mindmaps, audio, video, e-learning modules, LMS links, course resources, digital library links and learning activity tracking. SFRC's E-Content Development Cell is an existing institutional capability. ■cite■turn0search1■
H. Examination	Exam schedules, eligibility, attendance rule evaluation, CIA/external assessment placeholders, results, attainment views, CO/PO analytics and controlled COE functions. Official rules specify 68 days / 75% of 90 days per semester as the standard attendance threshold, with condonation provisions in specified shortage ranges. ■cite■turn1search8■
I. Mentoring	Tutor-ward assignment, mentee profile, semester personal record, academic/co-curricular records, meetings, goals, counselling notes, parent communication, follow-ups and progress. SFRC states each teacher is assigned 10–30 students and mentoring records are maintained. ■cite■turn0search0■
J. Library / IRC	Catalogue/search, availability, loans, due dates, fines, history, e-resources, e-books, e-dissertations, e-question papers, remote access links and library announcements. The current IRC page states 64,795 volumes and subscriptions including J-Gate, DELNET and INFLIBNET. ■cite■turn2search1■
K. Hostel	Hostel blocks, room allocation, occupants, warden, leave requests, parent/guardian approvals, special permission, rules, notices, facilities and maintenance complaints. Official site lists Priyadharshini, New Hostel and PG Hostel. ■cite■turn1search0■
L. Campus & Facilities	Buildings, classrooms, labs, auditorium/halls, sports, bus, canteen, cooperative store, day care, bank, ICT facilities and other support services. Official infrastructure pages list these facility categories. ■cite■turn0search7■

Module	Production scope
M. Campus Care / CivicFix	Complaint registration, image/location, category, AI classification, severity/priority, duplicate detection, assignment, SLA, escalation, resolution evidence, feedback and analytics.
N. Events & Announcements	Institution/department/cell events, date/time, venue, organizer, registration, reminders, attendance/participation, certificates, galleries, announcements and calendar. The official archive currently shows 629 entries and recent 2026 events across departments and cells. ■cite■turn0search4■
O. Career & Placement	Companies, drives, eligibility, applications, interview schedules, training, resume support, career guidance, higher studies and placement analytics.
P. Research	Research projects, publications, patents/copyrights, scholars, supervisors, research events, ethics/policy links and grant/activity records. The official site currently exposes a patents/copyright table. ■cite■turn1search2■
Q. Entrepreneurship	ACIDE, YWED, startup ideas, training, mentors, events, showcases and venture records. ACIDE lists a campus entrepreneurship ecosystem and startup examples including Kissan Innova, Yos Deliza, Kraftiga, Thukhil, Belle, ChemEmpor and Explora. ■cite■turn1search1■
R. Student Life & Outreach	Student Union, clubs, NSS, NCC, YRC, Red Ribbon Club, Social Service League, Citizen Consumer Club, Environmental Club, Financial Literacy Club, Agrarian Club, Self Defence Club, Human Rights Club, SWACHTA, UBA, etc. ■cite■turn1search4■turn1search9■
S. Alumnae	Profile, batch, department, career, mentoring, networking, events, opportunities and contributions; integration-ready for the existing Alumnae portal.
T. IQAC / Quality	AQAR, feedback, quality initiatives, action-taken items, academic attainment, evidence repository, best practices and dashboards.
U. Policies & Compliance	Code of conduct, anti-ragging, grievance pathways, IT policy, hostel rules, exam rules, research ethics and approved institutional documents.
V. Pragya AI	Permission-aware conversational assistant, live database queries, RAG over approved institutional documents, source-grounded answers, workflow initiation and safety controls.
W. Notifications	In-app notifications, email-ready events, optional web push through FCM, preferences and notification history.
X. Analytics	Attendance trends, academic performance, complaint SLA, event engagement, mentoring activity, library usage, placement activity and institutional KPIs.
Y. Audit & Security	Audit logs, role changes, sensitive-data access logs, complaint state transitions, admin actions, AI tool calls and source traceability.
Z. Settings & Configuration	Academic year, semesters, SLA policies, categories, priorities, notification rules, institutional content sources, feature flags and environment configuration.

6. CivicFix — Integrated as MyZone Campus Care

CivicFix is treated as a **major MyZone module**, not a separate application. User-facing naming can be “Campus Care” or “Maintenance & Complaints”, while CivicFix remains the internal product/module name.

The CivicFix project specification supplied in the project materials defines an AI-powered smart campus complaint and maintenance workflow. The core capability set is preserved below.

End-to-end workflow

Complaint → intake → AI understanding → category/severity/priority → responsible department/team → duplicate detection → intelligent assignment → SLA timer → escalation → work/update → resolution evidence → reporter confirmation/feedback → analytics/audit.

Capability	Implementation detail
Complaint intake	Description, category, location, image/evidence, reporter, timestamp, priority and optional contact details.
AI complaint understanding	Extract/normalize issue type, category, severity, urgency, likely responsible team and suggested action.
Duplicate detection	Compare new complaint against recent/open complaints using semantic similarity + structured filters such as location/category/status.

Capability	Implementation detail
Intelligent assignment	Route by category, location, priority, staff/team availability and current workload; keep assignment decision auditable.
SLA	Configurable target resolution times by priority; timer starts on accepted complaint; breach warning and escalation events are recorded.
Maintenance dashboard	Queue by status/priority/category/location/team, workload, aging, SLA compliance and resolution trends.
Resolution	Technician/staff update, status transitions, notes, attachments, resolution timestamp and optional before/after evidence.
Feedback	Reporter satisfaction, reopen request and resolution feedback.
AI query layer	Answer operational questions from actual complaint/maintenance data with permission checks.

Suggested demo categories grounded in SFRC facilities and IT/maintenance policies: Electrical, IT/Network, Computer, Projector, Classroom, Laboratory, Library, Hostel, Sports, Water, Furniture, Cleanliness, Infrastructure, Other. These are **MyZone configurable categories**, not claims that SFRC officially uses this exact taxonomy.

Suggested SLA values such as Critical 30 min, High 2 hr, Medium 8 hr and Low 24 hr may be used only as configurable demonstration defaults unless SFRC supplies an official SLA policy. They must never be presented as official institutional service commitments.

7. Pragya AI — Integrated Institutional Assistant

Pragya is an integrated AI layer, not the entire MyZone product. It must be **grounded, permission-aware and action-oriented**.

Question type	Data path	Example
Personal dynamic data	Authorized PostgreSQL query / API	"What is my attendance today?"
Institutional static knowledge	RAG → pgvector → approved source document	"What are library timings?"
Workflow request	AI intent → validated action → API	"Report the projector in Lab 2."
Cross-module request	Orchestrated authorized APIs	"What events are tomorrow and where?"
Sensitive request	RBAC/RLS check before retrieval	Never reveal another student's marks or private profile.

RAG pipeline

Approved website/document source → content extraction → chunking → metadata tagging → embeddings → pgvector → similarity search → context assembly → LLM → answer with source attribution/trace → audit log.

Dynamic data pipeline: user question → intent classification/tool selection → permission check → parameterized query/API call → result normalization → LLM response generation → no fabrication fallback.

Examples: "What is my timetable today?", "Am I eligible for semester exams?", "Who is my mentor?", "What are the library timings?", "How do I access e-resources?", "My lab projector is not working."

8. Production System Architecture

MYZONE – SFRC 360 | +-----+-----+ | | WEB EXPERIENCE AI / INTELLIGENCE | |
Next.js + TypeScript Pragya AI Tailwind + shadcn/ui | Responsive / PWA LLM + RAG + Tools | |
+-----+-----+ | FastAPI Backend |
+-----+-----+ | | Business APIs AI orchestration Integrations | |
(future ERMS etc.) +-----+-----+ | PostgreSQL / Supabase
+-----+-----+ | | Auth Storage pgvector | | RBAC/RLS Files/images RAG vectors

Recommended deployment shape

Frontend: Next.js production deployment (e.g., Vercel or equivalent). Backend: FastAPI on a production Python host. Database/Auth/Storage: Supabase/PostgreSQL. Source control: GitHub. AI provider: pluggable adapter so the

application is not hard-coded to a single LLM vendor.

For a fully free student/demo environment, use free tiers/open-source components wherever available and keep paid services optional. Production costs can arise from LLM inference, email/SMS, maps, high-volume storage and hosting beyond free quotas.

9. Technology Stack — Exact Role of Every Technology

Technology	Layer	What it does in MyZone
Next.js	Frontend application framework	Routing, layouts, server/client components, authenticated screens, responsive app shell, production web delivery.
TypeScript	Programming language	Typed domain models, API contracts, safer refactoring, maintainable frontend code.
React	UI component model	Reusable dashboard widgets, forms, tables, chat UI, cards, modals and feature components.
Tailwind CSS	Styling	Responsive design system, spacing, typography, states, dark mode and rapid UI development.
shadcn/ui	UI component foundation	Accessible dialogs, tables, forms, tabs, menus, sheets, dropdowns, command palette and consistent interaction patterns.
Lucide Icons	Icon system	Consistent product iconography.
React Hook Form	Form engine	Efficient forms for complaints, leave, events, profile updates, feedback and admin configuration.
Zod	Validation	Schema validation on form/API boundaries and typed parsing.
FastAPI	Backend/API framework	Business logic, REST APIs, AI endpoints, integrations, authorization checks and OpenAPI documentation.
Python	Backend/AI language	AI orchestration, RAG, NLP, data processing and backend services.
PostgreSQL	Relational database	Core institutional/operational data with strong relationships, constraints and transactions.
Supabase	Managed backend platform	PostgreSQL hosting, Auth, Storage, API tooling and database administration.
Supabase Auth	Authentication	Identity, sessions and login flows.
PostgreSQL RLS	Authorization enforcement	Row-level access control for student/parent/faculty/admin data.
Supabase Storage	Object storage	Complaint images, profile images, certificates, documents and approved knowledge files.
pgvector	Vector search	Semantic retrieval for Pragya RAG and CivicFix duplicate similarity.
LLM provider adapter	AI reasoning	Natural-language generation and tool-use; provider should be replaceable.
Recharts	Analytics charts	Attendance, academic, complaints, SLA, engagement and institutional KPI visualizations.
Firebase Cloud Messaging	Optional web push	Notifications and alerts; keep behind a notification abstraction.
Git + GitHub	Version control	History, rollback, branching, collaboration and deployment integration.
OpenAPI / Swagger	API documentation	Interactive API contract/testing for FastAPI.
Testing stack	Quality assurance	Unit, API, component and end-to-end tests; exact library can be selected by Antigravity within the architecture constraints.
Vercel / equivalent	Frontend hosting	Production delivery of Next.js.
Production Python host	Backend hosting	FastAPI deployment with environment secrets and health monitoring.

10. Module → Technology Mapping

Module	Primary technologies
Authentication & RBAC	Next.js, Supabase Auth, PostgreSQL RLS, FastAPI
Student/Faculty/Parent/Admin dashboards	Next.js, TypeScript, Tailwind, shadcn/ui, Recharts
Academics	Next.js, FastAPI, PostgreSQL/Supabase, Zod
Attendance / Marks / Results	Next.js, FastAPI, PostgreSQL, RLS
Timetable / Events	Next.js, PostgreSQL, FastAPI, calendar UI
Mentoring	Next.js, FastAPI, PostgreSQL, Storage, RLS
Library	Next.js, FastAPI, PostgreSQL, external resource links
Hostel	Next.js, FastAPI, PostgreSQL, Storage, notifications
Campus Care / CivicFix	Next.js, FastAPI, PostgreSQL, Storage, pgvector, AI provider
Duplicate complaint detection	FastAPI, embeddings/pgvector, PostgreSQL filters
Pragya AI	Next.js chat UI, FastAPI AI service, PostgreSQL tools, pgvector RAG, LLM adapter
RAG knowledge base	Python ingestion, pgvector, Supabase Storage, source metadata
Career / Placement	Next.js, FastAPI, PostgreSQL, notifications
Research	Next.js, FastAPI, PostgreSQL, Storage
Entrepreneurship	Next.js, FastAPI, PostgreSQL, Storage
Student Life / Outreach	Next.js, FastAPI, PostgreSQL, certificates in Storage
Admin Analytics	PostgreSQL queries/views, FastAPI, Recharts
Audit & compliance	PostgreSQL audit tables, FastAPI middleware, RLS
Notifications	FastAPI notification service, database preferences, optional FCM/email adapters

11. Core Database Model

Entity	Purpose
users	Identity record; auth UID, role/profile link, status, last login
roles / permissions	RBAC/capability matrix; avoids excessive top-level roles
students	Student profile, register number, programme, batch, semester, department
faculty	Faculty profile, department, designation, specialization
parents / guardians	Authorized relationship to student(s)
departments	Active department master; never hard-code current counts
programmes	Regular/SF/PG/PhD/certificate/diploma programme master
courses	Course code, title, credits, semester, programme mapping
enrollments	Student-course-semester relation
attendance_records	Date/session/course/student/attendance state
marks	Assessment component, marks, max marks, course/student
exams / results	Exam schedule, eligibility, result publication metadata
timetables	Day/time/course/faculty/room
assignments	Course, faculty, student/group, due date, submission
learning_resources	Videos, audio, mindmaps, modules, external links
mentorship_assignments	Mentor-mentee mapping, batch, start/end
mentoring_records	Meeting, goals, counselling, academic/co-curricular notes, follow-up
events	Title, category, department/cell, date/time, venue, organizer

Entity	Purpose
event_registrations	User/event relation, attendance/certificate
announcements	Audience, priority, publish window, attachment
library_items / loans	Catalogue, copy, loan, due date, return/fine
hostels / rooms / allocations	Hostel, block/room, occupancy, warden
hostel_leave_requests	Request, approver, parent/guardian, status
placements / companies / applications	Drive, eligibility, application and outcome
research_projects / publications / patents	Research records, outputs and metadata
clubs / organizations / memberships	Student-life and outreach participation
certificates	Event/course/activity certificate metadata + file
complaints	CivicFix case, reporter, location, category, severity, priority, status
complaint_assignments	Team/staff assignment, reason, workload snapshot
complaint_updates	Status updates, notes, attachments
sla_policies / sla_events	Configurable SLA targets, warnings, breaches, escalations
knowledge_sources / documents / chunks	RAG source registry, extracted content, embeddings and metadata
ai_conversations / messages	Pragya chat history with privacy controls
ai_tool_calls	Audit trail of AI actions/query tools
notifications	In-app/push/email-ready notification records
audit_logs	Security-sensitive and administrative action history
system_settings	Academic year, priorities, categories, feature flags and configuration

12. Security, Privacy & Production Controls

SFRC's IT policy emphasizes appropriate use of IT resources, ethical student-data retrieval, security/risk mitigation and periodic policy review. MyZone should therefore treat security as a first-class product feature rather than a final polish item. ■cite■turn0search6■

- **Authentication:** Supabase Auth; secure session handling; password reset; optional MFA later.
- **Authorization:** RBAC at application layer + PostgreSQL RLS as the database safety boundary.
- **Tenant/data scope:** A student can access only their own protected records; parents only their authorized ward(s); faculty only assigned/authorized academic and mentoring scopes.
- **AI security:** never pass unrestricted database dumps to the LLM; use explicit tools with permission checks and parameterized queries.
- **Prompt injection:** treat retrieved documents as untrusted content; system instructions and tool permissions remain authoritative.
- **File security:** MIME/type checks, size limits, private storage buckets for sensitive uploads, signed URLs where appropriate.
- **Audit:** record admin changes, permission changes, complaint transitions, AI tool calls and sensitive-data access.
- **Secrets:** environment variables; never commit API keys or service-role keys to GitHub.
- **API security:** validation, rate limiting for public/AI endpoints, structured errors, CORS policy, request IDs and health checks.
- **Database:** constraints, foreign keys, indexes, migrations and backup strategy.
- **Privacy:** minimum necessary data; retention policies; avoid exposing counselling/mentor notes or sensitive committee information to unauthorized roles.

13. UI / UX Product Direction

The product should look like a real institutional SaaS platform rather than a typical student project. The UI should use a consistent design system and responsive layout across student, faculty, parent and admin views.

UX principle	Implementation
Responsive	Desktop-first admin where useful; mobile-first student/parent flows; no separate mobile app required initially.
Navigation	Role-aware sidebar + mobile bottom/compact navigation for high-frequency student actions.
Dashboard	Actionable cards, alerts, upcoming events, quick actions and context-aware summaries.
Tables	Search, filter, sort, pagination, export where authorized.
Forms	Inline validation, clear errors, loading states, success states and unsaved-change protection where useful.
AI	Persistent Pragya entry point; citations/source cards; suggested prompts; clear distinction between information and action.
Campus Care	Simple complaint creation, visual status timeline, priority and SLA indicators.
Accessibility	Keyboard navigation, semantic controls, focus states, contrast and readable typography.
Trust	Show data freshness and source context for institutional information.

14. Backend API Organization

API domain	Representative routes
Auth / Profile	/auth/*, /users/me, /profiles/*
Students	/students/me, /students/{id}, /students/{id}/attendance, /students/{id}/marks
Academics	/departments, /programmes, /courses, /enrollments, /timetable
Mentoring	/mentoring/assignments, /mentoring/meetings, /mentoring/goals
Events	/events, /events/{id}, /events/{id}/register
Library	/library/items, /library/loans, /library/resources
Hostel	/hostels, /rooms, /hostel/leave-requests
CivicFix	/complaints, /complaints/{id}, /complaints/{id}/assign, /complaints/{id}/updates
Analytics	/analytics/student, /analytics/faculty, /analytics/admin
Pragya	/ai/chat, /ai/search, /ai/tools/*
Knowledge	/knowledge/sources, /knowledge/ingest, /knowledge/search
Admin	/admin/users, /admin/settings, /admin/audit

FastAPI should generate OpenAPI/Swagger documentation automatically. API response models should be explicit and versionable. Avoid exposing raw database models directly to the browser.

15. AI Data Architecture and Guardrails

Layer	Allowed role	Example
System prompts / policies	Developer-controlled	Security rules, tool boundaries, no fabrication policy
User context	Permission-scoped	Current user ID, role, programme, allowed scopes
Tool calls	Server-side only	get_my_attendance(), get_my_events(), create_complaint()
RAG context	Approved sources only	Exam rules, hostel rules, library info, IT policy, public institutional pages
LLM output	Constrained by application	Answer, ask clarification, propose action, or refuse unsupported request

Layer	Allowed role	Example
Audit	Server-side	Tool invocation, source IDs, response metadata

Pragya must never invent attendance, marks, exam eligibility, fees, policy wording or other institutional facts. If a source is missing or data is unavailable, the assistant should say so and direct the user to the appropriate institutional process.

16. How the Project Will Actually Be Built in Google Antigravity

Because the developer is using vibe coding and does not need to manually author every line, Antigravity should be treated as the implementation agent while the project owner controls requirements, review and acceptance.

Stage	Antigravity responsibility	Human responsibility
1. Repository bootstrap	Create Next.js + FastAPI structure, configs, linting, environment templates, Git setup.	Approve architecture and project structure.
2. Database	Create migrations/schema, seed scripts, indexes, RLS policies.	Verify data model and permissions.
3. Auth/RBAC	Implement login, roles, protected routes and server authorization.	Test each role's boundaries.
4. Core UI	Build design system, layout, dashboards, reusable components.	Review UX and consistency.
5. Modules	Implement one module at a time with API + DB + UI + validation + tests.	Accept/reject feature behavior.
6. AI	Implement Pragya tools, RAG pipeline, prompts, guardrails and audit.	Test factuality and privacy.
7. CivicFix	Implement complaint lifecycle, AI classification, duplicate detection, SLA and dashboard.	Test operational scenarios.
8. QA	Run typecheck, lint, tests, build, API checks and fix failures.	Perform end-user acceptance testing.
9. Deployment	Prepare production env, migrations and deployment configuration.	Configure accounts/secrets and approve release.

Golden rule for vibe coding: **Build** → **Run** → **Test** → **Fix** → **Commit** → **Continue**. Do not ask the agent to generate the entire enterprise platform in one uncontrolled pass. Use staged implementation and preserve a written architecture contract.

17. Three-Day Build Strategy

Day	Build target	Definition of done
Day 1 — Foundation	Repository, architecture, Supabase, database schema, seed data, Auth/RBAC, app shell, responsive design system, role dashboards.	All four main roles can sign in and reach correct protected dashboards; DB/RLS verified.
Day 2 — Product modules	Academics core, events, mentoring, library/hostel basics, Campus Care/CivicFix workflow, admin operations and analytics.	End-to-end demo flows work with realistic SFRC seed data.
Day 3 — AI + polish	Pragya, RAG, AI complaint classification/duplicate detection, SLA dashboard, notifications-ready layer, security review, testing, deployment and demo data.	Production-style UI, stable build, documented architecture, repeatable demo and deployment.

If time becomes constrained, do not remove the architecture. Reduce breadth of secondary modules while keeping the core: Auth/RBAC + Dashboard + Academics + Events + CivicFix + Pragya + Admin analytics. Secondary modules can be represented as integration-ready foundations.

18. Testing & Quality Gates

Test layer	Examples
Type / static	TypeScript typecheck, Python type checks where configured, linting.
Unit	Priority calculation, attendance eligibility logic, SLA timers, duplicate score thresholds, permission helpers.
API	Auth, student access boundaries, complaint creation/assignment, admin endpoints.
RLS/security	Student A cannot read Student B; parent can read only linked ward; faculty access follows assignment scope.
UI/component	Forms, loading/error states, tables, dialogs, responsive layouts.
E2E	Login → dashboard → attendance; login → complaint → assignment → resolution; Pragya → database answer.
AI evaluation	Groundedness, refusal on unsupported facts, privacy leakage tests, prompt-injection tests, tool-call correctness.
Build/deploy	Clean install, environment validation, migration run, production build, health endpoint.

19. Realistic SFRC Seed Data Plan

Use official public data as seed/reference content, but do not pretend that private student/parent/faculty records came from the public website. Synthetic demo identities should be clearly marked as demo data.

- **Institution:** name, address, contact, history, milestones and approved public policies.
- **Academics:** current programme lists from regular/self-financing/certificate/diploma pages.
- **Departments:** import current active department records from the official departments directory; derive counts dynamically.
- **Faculty:** use public faculty profiles where required for institutional display; do not expose private data.
- **Mentoring:** seed current public mentor list structure by department and batch where appropriate; the official mentoring page publishes department/batch mentor lists. `■cite■turn0search0■`
- **Events:** seed a selected recent sample from the official archive for demo; do not bulk-copy all 629 records unless authorized.
- **Library:** seed resource categories and e-resource links; do not fabricate individual live loan records.
- **Hostel:** seed buildings, rules and facility descriptions; use synthetic room allocations for demos.
- **Policies:** ingest approved public policy pages/PDFs into Pragya RAG with source URLs and crawl dates.
- **CivicFix:** synthetic complaint dataset covering electrical, network, projector, classroom, lab, library, hostel and infrastructure cases.
- **Academic private data:** synthetic attendance/marks/timetable for demonstration unless authorized data is provided.

20. Integration Strategy — Current vs Future

System / source	Now	Future
SFRC official website	Public content + approved RAG ingestion / reference data	Automated controlled content refresh if permitted
SFRC ERMS	No unverified direct integration; MyZone has its own schema	Authorized API / data exchange adapter
Student/Parent portals	No password-sharing or scraping	Official integration if institution authorizes it
Library resources	Public resource links + MyZone catalogue model	ILS/API integration if available
Events archive	Seed selected public events	Approved synchronization feed
Alumnae portal	MyZone module / link	Authorized integration
FCM	Optional notification channel	Production push notifications

The architecture should use an **integration adapter layer** so an external system can be added later without rewriting the core product. Never store external system credentials in frontend code.

21. Production Readiness Checklist

- 1. Environment separation: local / staging / production.
- 2. Secrets stored only in environment/secret management.
- 3. Database migrations committed and repeatable.
- 4. RLS policies enabled and tested.
- 5. Role/capability matrix documented.
- 6. API validation and error handling implemented.
- 7. File upload restrictions implemented.
- 8. Audit logs enabled for sensitive/admin actions.
- 9. AI tool calls permission-checked and logged.
- 10. RAG sources have metadata and source URLs.
- 11. Unsupported AI answers fall back safely.
- 12. Indexes created for high-use queries.
- 13. Pagination used for large tables.
- 14. Loading, empty and error states present.
- 15. Responsive UI tested on mobile and desktop.
- 16. Automated tests run before deployment.
- 17. Production build succeeds from a clean environment.
- 18. Backup/recovery plan documented for database.
- 19. Monitoring/health endpoint configured.
- 20. Demo data separated from real institutional data.
- 21. ERMS integration claims explicitly marked as future/authorized-only.

22. Final Deliverables Expected

Deliverable	Expected output
Source repository	GitHub repository with frontend/backend/database migration/seed code.
Web application	Responsive MyZone application with role-aware dashboards.
API	FastAPI service with OpenAPI documentation.
Database	PostgreSQL schema, migrations, RLS and seed scripts.
AI	Pragya chat + database tools + RAG + audit controls.
CivicFix	Complaint workflow + AI triage + duplicate detection + SLA + maintenance dashboard.
Admin	User/configuration/analytics/audit functions.
Documentation	README, architecture, environment setup, database ERD, API docs, security notes and deployment guide.
Demo	Repeatable scenario-based demo with synthetic data.

23. How to Present MyZone as a Production-Level Product

Do not present it as: “a college management system made for a final-year project.”

Present it as: “MyZone – SFRC 360 is a unified smart-campus experience and institutional intelligence platform designed around SFRC’s existing digital ecosystem, with an AI assistant and an AI-powered campus maintenance/complaint workflow.”

The strongest technical story is the combination of: (1) real institutional requirements, (2) relational data architecture, (3) role-aware security, (4) RAG + tool-based AI, (5) AI-assisted maintenance operations, (6) analytics and auditability, and (7) integration readiness with existing ERMS rather than pretending to replace it.

Core demo narrative

Student logs in → sees personalized dashboard → asks Pragya about attendance/timetable → registers for an event → reports a projector issue → CivicFix classifies and routes it → maintenance staff updates status → SLA dashboard changes → student receives update → admin sees operational analytics. This single narrative demonstrates frontend, backend, database, RBAC, AI, RAG, workflow automation, notifications and analytics in one coherent product story.

24. Primary Sources & Reference Notes

Source	Reference
SFRC official homepage	https://www.sfrcollege.edu.in/
SFRC Best Practices / ERMS	https://www.sfrcollege.edu.in/box-link/best-practices
SFRC Mentoring System	https://www.sfrcollege.edu.in/box-link/mentoring-system
SFRC Regular Programmes	https://www.sfrcollege.edu.in/programmes/regular
SFRC Self-Financing Programmes	https://www.sfrcollege.edu.in/programmes/self-financing
SFRC Certificate & Diploma Courses	https://www.sfrcollege.edu.in/programmes/certificate-and-diploma-courses
SFRC Examination Rules & Regulations	https://www.sfrcollege.edu.in/examination/rules-and-regulations
SFRC Maintenance Policy	https://www.sfrcollege.edu.in/infrastructure/maintenance-policy
SFRC IT Policy	https://www.sfrcollege.edu.in/infrastructure/it-policy
SFRC Hostel	https://www.sfrcollege.edu.in/infrastructure/hostel
SFRC Other Infrastructure	https://www.sfrcollege.edu.in/infrastructure/other-infrastructure
SFRC Information Resource Centre	https://www.sfrcollege.edu.in/information-resource-centre/about-irc
SFRC Events Archive	https://www.sfrcollege.edu.in/archives
SFRC Research & Innovations	https://www.sfrcollege.edu.in/research/research-innovations
SFRC ACIDE	https://www.sfrcollege.edu.in/entrepreneurship/acide
SFRC YWED	https://www.sfrcollege.edu.in/entrepreneurship/ywed
SFRC NSS / Outreach	https://www.sfrcollege.edu.in/social-awareness-programme/national-service-scheme
SFRC NCC	https://www.sfrcollege.edu.in/social-awareness-programme/national-cadet-corps
SFRC Youth Red Cross	https://www.sfrcollege.edu.in/social-awareness-programme/youth-red-cross
SFRC Computer Science Department	https://www.sfrcollege.edu.in/department/computer-science
SFRC Milestones	https://www.sfrcollege.edu.in/about-us/milestones-of-sfrc
CivicFix project specification	Project material supplied earlier in the MyZone project conversation; integrated here as the Campus Care/CivicFix requirements baseline.

Web-source snapshot date: 27 September 2026. Website content is live and can change. Historical figures and historical programme names should not be treated as current without a dated source.

This specification intentionally distinguishes official SFRC facts from MyZone design decisions, configurable demo values, and future integration assumptions.

Appendix A — Recommended Repository Structure

```
myzone-sfrc360/
├── apps/
│   ├── web/ # Next.js + TypeScript
│   └── app/
│       ├── components/
│       ├── features/
│       ├── lib/
│       ├── styles/
│       └── api/ # FastAPI
├── core/
│   ├── models/
│   ├── schemas/
│   ├── services/
│   ├── ai/
│   ├── integrations/
│   ├── tests/
│   ├── supabase/
│   ├── migrations/
│   ├── seed/
│   ├── functions/
│   ├── knowledge/
│   ├── sources/
│   ├── processed/
│   ├── metadata/
│   ├── docs/
│   ├── architecture/
│   ├── api/
│   ├── security/
│   └── deployment/
├── scripts/
├── .env.example
├── README.md
└── docker-compose.yml # local development where useful
```

Antigravity should preserve this separation: UI logic in the web app, business/API logic in FastAPI, persistence and security in PostgreSQL/Supabase, AI orchestration in a dedicated service layer, and external integrations behind adapters.

Appendix B — Minimum Demo Scenarios

Scenario	Flow
Student	Login → dashboard → attendance → timetable → Pragma question → event registration.
Parent	Login → ward profile → attendance/progress → mentor/notice view.
Faculty	Login → assigned course → attendance/marks → mentee list → mentoring record.
Admin	Login → institution dashboard → complaint queue → SLA analytics → user/configuration view.
CivicFix	Student creates "projector not working" → AI category/priority → duplicate check → assignment → status update → resolution → feedback.
Pragma RAG	Ask library/hostel/exam-policy question → retrieve approved source → answer with source context.
Security	Attempt cross-student data access → RLS/API authorization blocks it.

END OF SPECIFICATION